

Exam: Data Analysis and Retrieval 2024

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General guidelines

- You are allowed to have one A4 of useful formulas that you can use during the exam.
- All questions are compulsory.
- When answering the questions, please write the intermediate steps of the derivations, not just the end numerical results.

Question 1

You have a collection of three books (Book1, Book2, Book3), each book have the theme "Action" or "Drama", see Table below. You want to add a fourth book for your collection (Book4). You want to classify this book either as an "Action" or as a "Drama" book using the words in Book1, Book2 and Book3 as training examples.

	words in the book	type of book
Book1	run tears jump jump	Action
Book2	look jump stay	Drama
Book3	tears sun moon	Drama
Book4	jump look sun	?

a) Determine $\hat{P}(\text{Action}|\text{Book4})$ and $\hat{P}(\text{Drama}|\text{Book4})$ using the multinomial Naive Bayes model. Use Laplace smoothing. To which class is Book4 assigned to when using this model?

Solution:

$$\hat{P}(\text{Action}|\text{Book4}) \cong P(\text{Action}|\text{jump look sun}) \quad (1)$$

$$= \hat{P}(\text{Action}) \cdot \hat{P}(\text{jump}|\text{Action}) \cdot \hat{P}(\text{look}|\text{Action}) \cdot \hat{P}(\text{sun}|\text{Action}) \quad (2)$$

$$= \frac{1}{3} \cdot \frac{3}{11} \cdot \frac{1}{11} \cdot \frac{1}{11} = \left(\frac{1}{11}\right)^3 = \frac{1}{2197} = 0.000455 \quad (3)$$

$$\hat{P}(\text{Drama}|\text{Book4}) \cong P(\text{Drama}|\text{jump look sun}) \quad (4)$$

$$= \hat{P}(\text{Drama}) \cdot \hat{P}(\text{jump}|\text{Drama}) \cdot \hat{P}(\text{look}|\text{Drama}) \cdot P(\text{sun}|\text{Drama}) \quad (5)$$

$$= \frac{2}{3} \cdot \frac{2}{13} \cdot \frac{2}{13} \cdot \frac{2}{13} = \frac{2}{3} \cdot \frac{8}{13^3} \quad (6)$$

$$= \frac{16}{3 \cdot 13^3} = \frac{16}{6591} = 0.00242 \quad (7)$$

$\hat{P}(\text{Drama}|\text{Book4}) > \hat{P}(\text{Action}|\text{Book4})$ so the book is classified as Drama.

Question 2

After several years of giving the DAR course, the lecturer is analyzing the potential grades that students could get at the exam based on the number of days the students reported spending for the preparation of the exam. The lecturer has the following data:

Based on these data, the lecturer estimates the following linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2),$$

Number of days student prepared the exam (x_i)	0	1	2	3	4
Grade of the student (y_i)	5	7	8	9	10

using the least squares method.

- a) Determine the least squared estimators β_0 and β_1 .
- b) Determine how many extra points does a student secure for every extra day of exam preparation?
- c) The p-value obtained for coefficient β_1 is 0.0019. Provide an interpretation of this p-value.

Solution:

a)

$$b = [\beta_0, \beta_1]^T \quad (8)$$

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \quad (9)$$

$$y = \begin{bmatrix} 5 \\ 7 \\ 8 \\ 9 \\ 10 \end{bmatrix} \quad (10)$$

$$b = (X^T X)^{-1} X^T Y \quad (11)$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix} \quad (12)$$

$$(X^T X)^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{150 - 100} \begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix} \quad (13)$$

$$X^T \cdot Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \\ 8 \\ 9 \\ 10 \end{bmatrix} = \begin{bmatrix} 5 + 7 + 8 + 9 + 10 \\ 7 + 16 + 27 + 40 \end{bmatrix} = \begin{bmatrix} 39 \\ 90 \end{bmatrix} \quad (14)$$

$$b = [\beta_0 \quad \beta_1] = \frac{1}{50} \begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix} \cdot \begin{bmatrix} 39 \\ 90 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 1170 - 900 \\ -390 + 450 \end{bmatrix} \quad (15)$$

$$b = \begin{bmatrix} 270 \\ 60 \end{bmatrix} \cdot \frac{1}{100} \quad (16)$$

$$b = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 5.4 \\ 1.2 \end{bmatrix} \quad (17)$$

b) $\beta_1 = 1.2$ extra pointsc) The ($H_0 : \beta_1$ is zero) is rejected and thus the coefficient β_1 is statistically significant.

Question 3

An airline is investigating the impact of the price of the flight tickets from Amsterdam to Tokyo on people's willingness to purchase a ticket. Based on the data from 100 respondents, the airline obtained the following logistic regression model:

$$P(\text{person purchases flight ticket} \mid \text{price of ticket}) = P(Y = 1 \mid X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}},$$

with X the price of the flight ticket.

Based on the data, the airline has determined that $\beta_0 = 0.1$ and $\beta_1 = -0.2$.

a) The airline wants to increase the price of the flight tickets from Amsterdam to Tokyo from 1K euros (1.000 euros) to 1.5K euros (1.500 euros). Determine the increase/decrease in the probability that a person will purchase a flight ticket when an increase from 1k euros to 1.5k euros is applied.

b) Interpret the values obtained for coefficients β_0 and β_1 .

Solution:

a)

$$P(Y = 1 \mid X = 1) = \frac{e^{0.1 - 0.2 \cdot 1}}{1 + e^{0.1 - 0.2 \cdot 1}} = 0.475 \quad (18)$$

$$P(Y = 1 \mid X = 1.5) = \frac{e^{0.1 - 0.2 \cdot 1.5}}{1 + e^{0.1 - 0.2 \cdot 1.5}} = 0.45 \quad (19)$$

When a 500 euros increase is applied, the probability for a person to purchase a ticket decreases from 0.475 to 0.45, *i.e.*, a decrease of 2.5 percentage points.

OR

$$P(Y = 1 \mid X = 1) - P(Y = 1 \mid X = 1.5) = \frac{e^{0.1 - 0.2 \cdot 1}}{1 + e^{0.1 - 0.2 \cdot 1}} - \frac{e^{0.1 - 0.2 \cdot 1.5}}{1 + e^{0.1 - 0.2 \cdot 1.5}} \quad (20)$$

b) When $X = 0$, so the flight ticket is free, the probability that someone gets this ticket is

$$\frac{e^{\beta_0}}{1 + e^{\beta_0}} = \frac{e^{0.1}}{1 + e^{0.1}} = 0.52 \quad (21)$$

β_0 is positive, so when the ticket is free, there is a preference to take the ticket. Also, $\beta_1 = -0.2$, so an increase in the flight ticket price will lead to a decrease in the probability that this will be purchased.

A one unit increase in the price is associated with an increase in the log odds of the price by -0.2 or equivalently a decrease in the odds of purchasing the ticket by $e^{-0.2} = 0.818$.

Question 4

Consider the co-occurrence matrix below, i.e., raw counts of the number of times the target word and the context word co-occur in the training corpus.

		context		
		television	holidays	outdoor
target word	student	1	20	30
	writer	20	30	10
	freelancer	10	1	3

a) Determine the cosine between the target word "student" and the content word "freelancer".

Solution:

$$\text{cosine}(\text{student}, \text{freelancer}) = \frac{1 \cdot 10 + 20 \cdot 1 + 30 \cdot 3}{\sqrt{1 + 20^2 + 30^2}} = 0.0033 \quad (22)$$

Question 5 - TODO

Consider the following regression equation for a problem

$$E[y|x_1, x_2] = \beta_0 x_1 + \beta_1^2 x_2.$$

You aim to estimate the unknown coefficients β_0 and β_1 using the least squares method based on a set of n observations $\{(y_i, x_{1,i}, x_{2,i})\}, i = 1, \dots, n$.

- Specify the residual sum of squares (RSS) for this regression problem.
- You have available the following data points:

y	x_1	x_2
1	0	1
2	1	1

You used the gradient descent algorithm, having $\eta = 0.01$, and after several steps you obtained $b^t = (1, 2)$, i.e., $\beta_0^t = 1, \beta_1^t = 2$.

Calculate b^{t+1} using the method of gradient descent.

Solution:

a)

$$RSS(\beta_0, \beta_1) = \sum_{i=1}^m (y_i - \beta_0 \cdot x_{1,i} - \beta_1^2 \cdot x_{2,i})^2 \quad (23)$$

b)

$$\frac{\delta RSS_i}{\delta \beta_0} = \sum_{i=1}^m 2(y_i - \beta_0 x_{1i} - \beta_1 x_{2i}) \cdot (-x_{1i}) \quad (24)$$

$$\frac{\delta RSS_i}{\delta \beta_1} = \sum_{i=1}^m 2(y_i - \beta_0 x_{1i} - \beta_1^2 x_{2i}) \cdot (-2\beta_1 x_{2i}) \quad (25)$$

$$\frac{\delta RSS_i}{\delta \beta_0} = 2(y_1 - \beta_0 x_{11} - \beta_1^2 x_{21}) \cdot (-x_{11}) + 2(y_2 - \beta_0 x_{12} - \beta_1^2 x_{22}) \cdot (-x_{12}) \quad (26)$$

$$\frac{\delta RSS_i}{\delta \beta_1} = 2(y_1 - \beta_0 x_{11} - \beta_1^2 x_{21}) \cdot (-2\beta_1 x_{21}) + 2(y_2 - \beta_0 x_{12} - \beta_1^2 x_{22}) \cdot (-2\beta_1 x_{22}) \quad (27)$$

$$\frac{\delta RSS_i}{\delta \beta_0} = 2(1 - 1 \cdot 0 - 2^2 \cdot 1) \cdot (-0) + 2(2 - 1 \cdot 1 - 2^2 \cdot 1) \cdot (-1) = 0 + 6 = 6 \quad (28)$$

$$\frac{\delta RSS_i}{\delta \beta_1} = 2(1 - 1 \cdot 0 - 2^2 \cdot 1) \cdot (-2 \cdot 2 \cdot 1) + 2(2 - 1 \cdot 1 - 2^2 \cdot 1) \cdot (-2 \cdot 2 \cdot 1) = 24 + 24 = 48 \quad (29)$$

$$\nabla RSS = \begin{bmatrix} 6 \\ 48 \end{bmatrix} \quad (30)$$

$$\eta = \frac{1}{100} \quad (31)$$

$$\beta_0^{t+1} = \beta_0^{(t)} - \eta \frac{\delta RSS}{\delta \beta_0} \quad (32)$$

$$\beta_1^{t+1} = \beta_1^{(t)} - \eta \frac{\delta RSS}{\delta \beta_1} \quad (33)$$

$$\Rightarrow \beta_0^{t+1} = 1 - \frac{1}{100} \cdot 6 = 1 - 0.06 = 0.94 \quad (34)$$

$$\Rightarrow \beta_1^{t+1} = 2 - \frac{1}{100} \cdot 48 = 2 - 0.48 = 1.52 \quad (35)$$

Question 6

A holiday resort is ranked by the tourists that visited it as 1: Excellent, 2: Good, 3: Bad. The tourists consider: i) the Cleanliness level of the accommodation, which takes values $\{1, 2, 3, 4, 5\}$ with value 5 indicating the highest level of Cleanliness, and ii) the Distance to the City Center (in km). The manager of the resort developed a proportional odds logistic regression model to estimate the probability that a tourist ranks the resort as 1:Excellent. Based on the reviews from the tourists, the manager estimated:

Coefficient for	Estimate
Cleanliness	-0.65
Distance to City Center	0.78

Threshold	Estimate
t_1	1
t_2	2

a) Determine the probability that a tourist that evaluates the Cleanliness of the accommodation to be 2 (Good) and the Distance to the City Center to be 4, will rank the resort as 1 (Excellent).

b) For a level of Cleanliness of 2, what is the maximum Distance to the City Center that a resort should be located at such that the probability of ranking this resort as 1: Excellent is at least 50%?

Solution:

a)

$$P(y \leq 1 | \text{Clean} = 2, \text{Distance} = 5) \quad (36)$$

$$= P(y \leq 1 | t_1 - \beta_1 x_{\text{Clean}} - \beta_2 x_{\text{Distance}}) \quad (37)$$

$$= \frac{e^{1+0.65 \cdot 2 - 0.78 \cdot 5}}{1 + e^{1+0.65 \cdot 2 - 0.78 \cdot 5}} = 0.30 \quad (38)$$

$$y = \begin{cases} 1 & \text{if } y^* \leq 1 \\ 2 & \text{if } 1 < y^* \leq 2 \\ 3 & \text{if } 2 < y^* \end{cases}$$

b)

$$P(y = 1) = P(y \leq 1) = F(t_1 - \beta_0^T X_i) \geq 0.5 \quad (39)$$

$$= \frac{e^{1+0.65 \cdot 2 - 0.78 \cdot X_2}}{1 + e^{1+0.65 \cdot 2 - 0.78 \cdot X_2}} \geq 0.5 \quad (40)$$

Solving for X_2 : $X_2 \leq \frac{115}{39} \iff X_2 \leq 2.94$.

So the resort should be at most 2.92 km away from the city center; or using integers $X_2 = 2$ km away from the city center at most.

$$e^{1+0.65 \cdot 2 - 0.78 \cdot X_2} \geq \frac{1}{2} + \frac{1}{2} e^{1+0.65 \cdot 2 - 0.78 \cdot X_2} \quad (41)$$

$$\frac{1}{2} e^{1+0.65 \cdot 2 - 0.78 \cdot X_2} \geq \frac{1}{2} \quad (42)$$

$$e^{1+0.65 \cdot 2 - 0.78 \cdot X_2} \geq 1 \iff 1 + 0.65 \cdot 2 - 0.78 \cdot X_2 \geq 0 \quad (43)$$

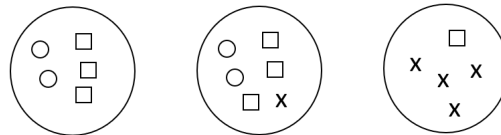
$$\iff -0.78 \cdot X_2 \geq -2.3 \quad (44)$$

$$X_2 \leq \frac{2.3}{0.78} \quad (45)$$

$$X_2 \leq 2.95 \quad (46)$$

Question 7

After several iterations of the K-means algorithm, you obtain the following clustering.



a) Calculate the Rand Index of this clustering.

Solution:

$$TP + FP + FN + TN = \binom{16}{2} = 20$$

$$TP = \binom{3}{2} + \binom{2}{2} + \binom{2}{2} + \binom{3}{2} + \binom{4}{2} = 14$$

$$TP + FP = \binom{5}{2} + \binom{6}{2} + \binom{5}{2} = 21$$

$$FN = 4 + (9 + 3) + (3 + 4) = 23$$

$$FN + TN = 120 - (TP + FP) = 120 - 35 = 85$$

$$TN = 85 - FN = 85 - 23 = 62$$

$$RI = \frac{14+62}{120} = 0.63$$

Question 8 - DONE

Consider a discrete random variable X that takes values $\{1, 2, 3\}$ with equal probability. You conduct a simulation of X and observe the following data:

$\{1, 2, 2, 2, 3, 1, 2, 3, 2\}$.

a) Determine a 95% confidence interval for X .

Solution:

$$E[X] = \frac{18}{9} = 2$$

$$Var[X] = \frac{\sum_i (x_i - E[X])^2}{n-1} = 0.5$$

$$CI : 2 \pm 1.96 * \frac{\sqrt{0.5}}{\sqrt{9}}$$